



Visual Editor vs JSON Editor in MongoDB Atlas

Two powerful ways to define how MongoDB organizes and searches your data — each built for a different kind of user.

- ① **First, understand:** An **Index Definition** is a set of instructions you give MongoDB — telling it which field contains embeddings, which field is searchable, the similarity type, metadata fields, and dimensions. MongoDB needs these instructions to function correctly.

Visual Editor

Click dropdowns, buttons, and forms. No JSON knowledge needed.

{ } JSON Editor

Write index configuration directly. Full control, full power.

The Visual Editor

METHOD 1

The Visual Editor is a **UI-based configuration tool**. Instead of writing JSON manually, you interact with a graphical interface — clicking dropdowns, buttons, checkboxes, and forms to define your index.

Simple Analogy

Think of it like filling out a **Google Form online**. You select values and click options — no coding required whatsoever.

Who It's For

Beginners, demo builders, and anyone who wants **quick setup** without needing to know JSON syntax.

What It Does Internally

Behind the scenes, the Visual Editor **secretly generates JSON** for you — you just never see it.

Visual Editor: How It Works

Example: Setting Up Vector Search

The Visual Editor asks you a series of simple questions. You fill in the answers:

01

Field Name?

You enter: embedding

02

Dimensions?

You enter: 1536

03

Similarity Type?

You enter: cosine

MongoDB then **internally creates the JSON automatically**. You never write a single line of code.

```
{ "fields": [  
  {  
    "type": "vector",  
    "path": "embedding",  
    "numDimensions": 1536,  
    "similarity": "cosine"  
  }  
]}
```

Advantages

- Beginner-friendly
- Quick setup
- No JSON knowledge needed
- Easy for demos
- Useful for POC (Proof of Concept)

Limitations

The Visual Editor only exposes **common and simple configurations** — not all advanced options. A UI simply cannot surface every possible complex configuration easily.

The JSON Editor

METHOD 2

The JSON Editor is a **manual configuration mode**. You directly write the index configuration JSON yourself — giving you complete, unrestricted control over every setting.

The Analogy

Visual Editor = Drag-and-drop website builder

JSON Editor = Writing HTML/CSS manually

Both create the same result — but manual gives more control.

Another Analogy

Visual Editor = Ordering a burger through a food delivery app UI

JSON Editor = Giving exact recipe instructions directly to the chef

Example: Vector Search + Metadata Filtering

```
{ "fields": [  
  {  
    "type": "vector",  
    "path": "embedding",  
    "numDimensions": 1536,  
    "similarity": "cosine"  
  },  
  {  
    "type": "filter",  
    "path": "module"  
  },  
  {  
    "type": "filter",  
    "path": "priority"  
  }  
]}
```

With this single JSON definition, you configure both **vector search** and **metadata filtering** simultaneously — something the Visual Editor cannot easily expose.

Why JSON Editor Wins in Production

The Most Important Difference

Visual Editor

MongoDB controls the available options. You work within the UI's boundaries.

JSON Editor

YOU control everything. No limits. No UI constraints. Full power.

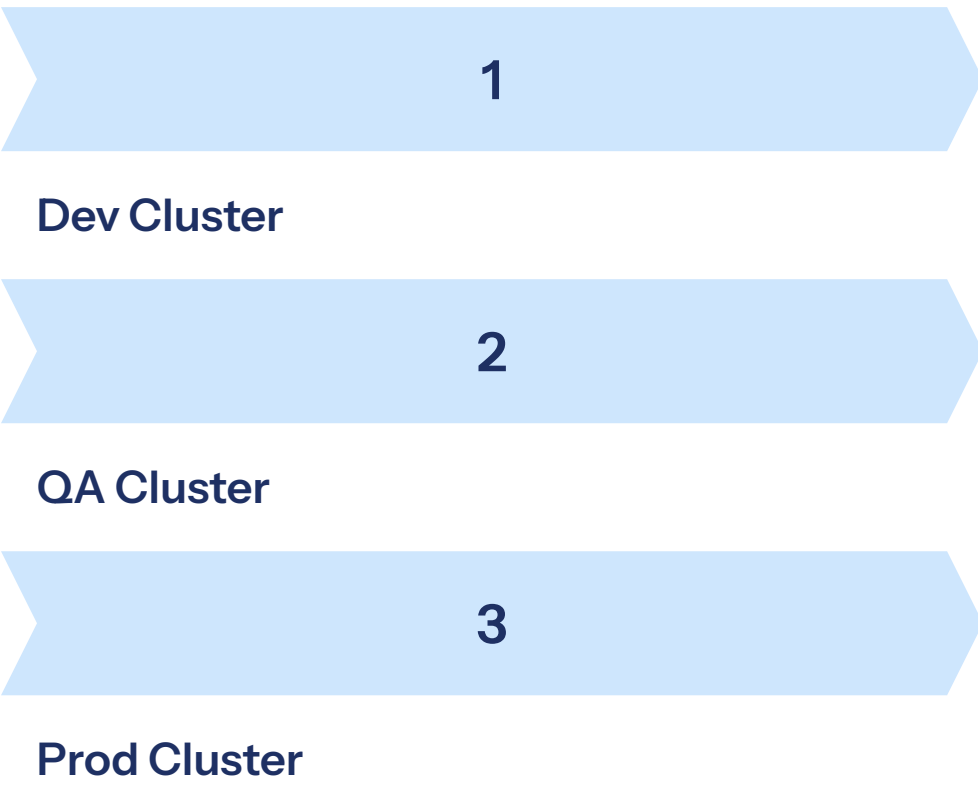
Enterprise Real Usage

In real companies, **nobody manually clicks the UI repeatedly**. Instead, JSON definitions are stored in version control and deployed automatically:

- GitHub / GitLab repositories
- CI/CD pipelines
- Infrastructure-as-code workflows

Reusability Across Environments

Suppose you have three environments. Instead of manually creating indexes three times, the **same JSON is reused everywhere**:



One file — `vector-index.json` — deploys automatically to all three. No repeated manual work.

Why Advanced Users Prefer JSON



Advanced Config

Access every possible configuration option, not just what the UI exposes.



Git Versioning

Track changes, roll back, and collaborate using standard version control.



Automation

Integrate with CI/CD pipelines for fully automated deployments.



Reusability

Write once, deploy everywhere — Dev, QA, and Prod from a single source.

 **Key Insight:** The Visual Editor is **NOT a different technology**. It is simply a UI wrapper around the same JSON configuration. Both tools produce identical results — the difference is only in who writes the JSON: you or the UI.

 Visual Editor

Easy graphical UI for creating **simple and common** index configurations. Perfect for beginners, demos, and POCs.

 JSON Editor

Direct low-level configuration providing **full control**, advanced customization, automation, reusability, and production-grade flexibility.